

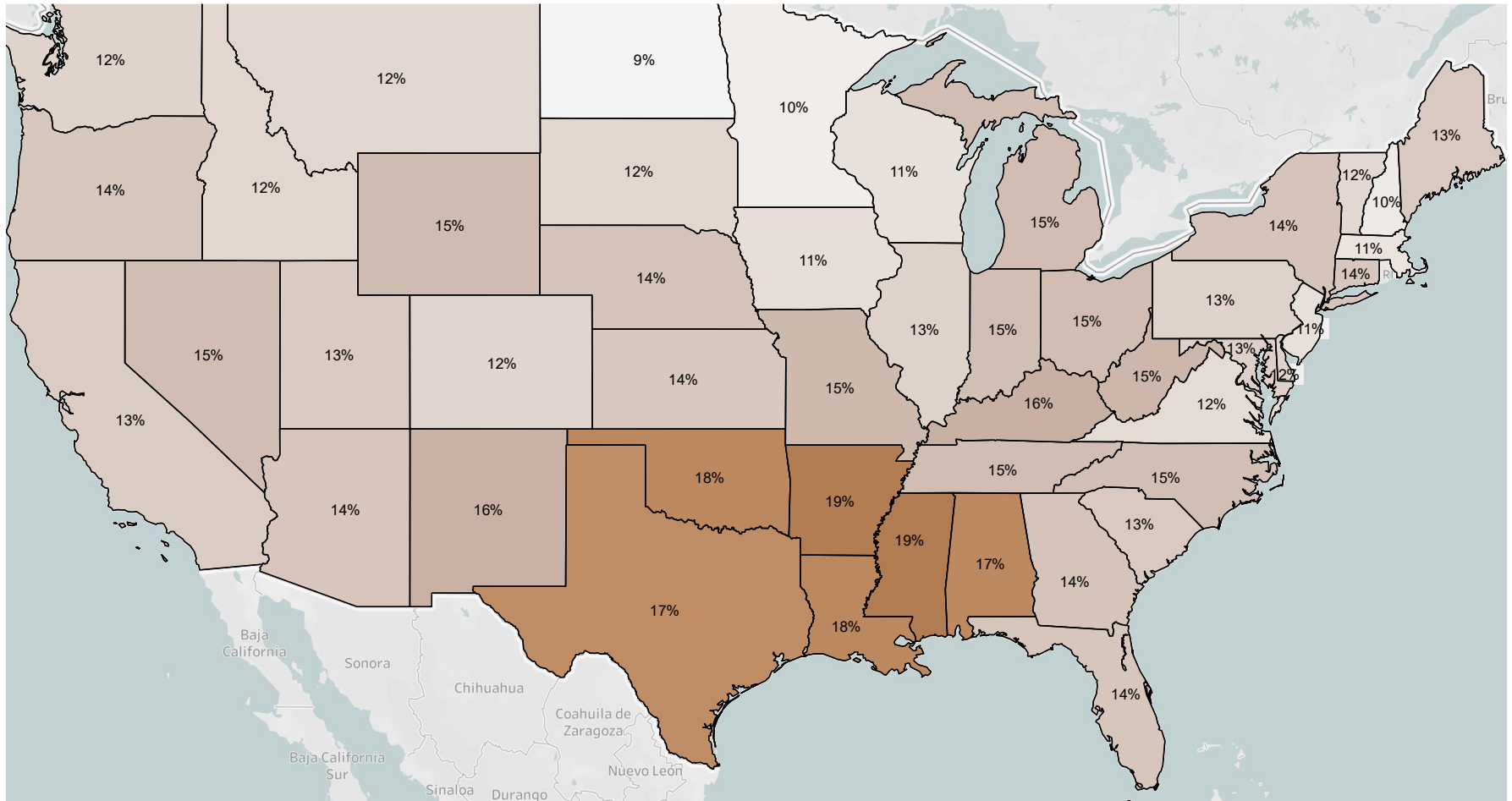
American Food Insecurity: No Way Out

Data Visualization Final Presentation

Millions of Americans Don't Have Enough Food

Despite a \$32 trillion GDP, food inequality in America remains deeply entrenched. Over 45 million people, across 18 million households, struggled to afford enough food in 2024 — this included 14.1 million children. The burden falls hardest on the South, where rates climb are nearly double those in lowest-insecurity states.

American food inequality by state



Research Framework

Aims

Analyze the relationship between food insecurity, poverty, and health outcomes across U.S. counties

Audience

Policymakers, researchers, and stakeholders interested in food security and public health policy

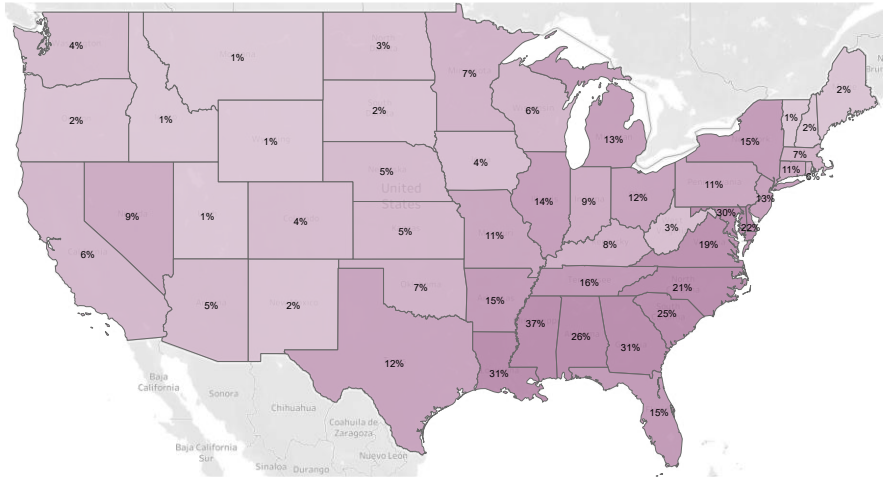
Methodology

Data analysis of county-level statistics from Feeding America, the CDC, Food Research Atlas and Census sources

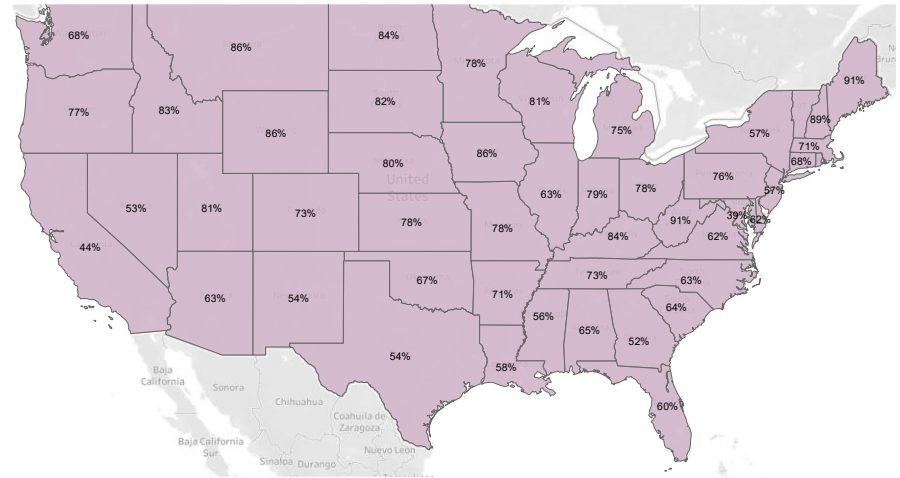
| The Problem is Driven by an Ethnic Divide

Black and Hispanic households face food insecurity at rates nearly double that of White households. In the South, where these communities exceed 30% of the population, this disparity compounds into a regional crisis.

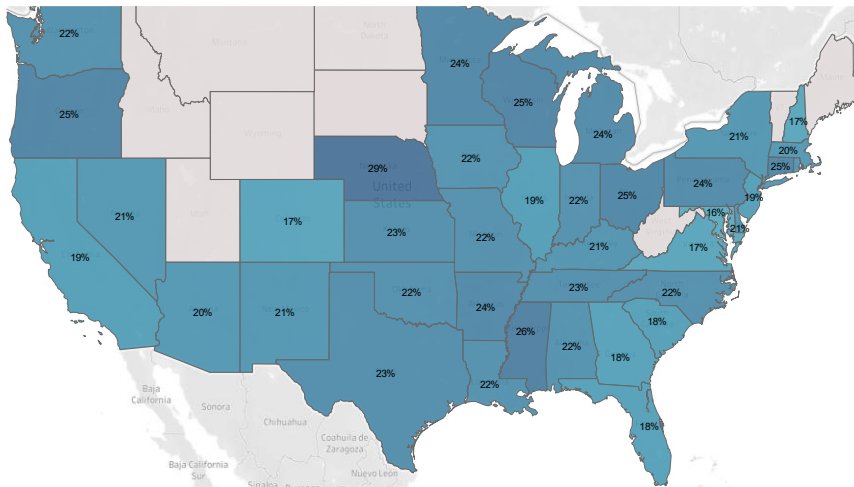
Southern states are dominated by Black and Hispanic communities...



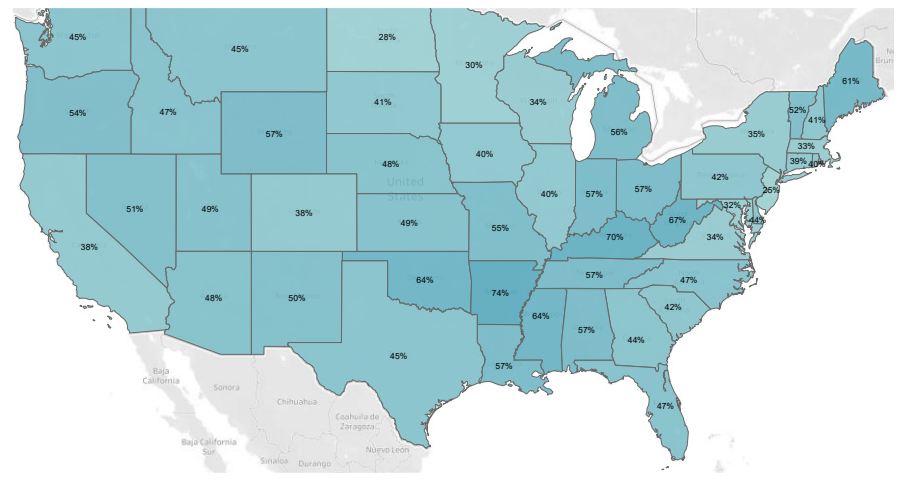
...while White populations are equally distributed across America



Across America, food insecurity is worse for Black and Hispanic populations...



...compared to White populations

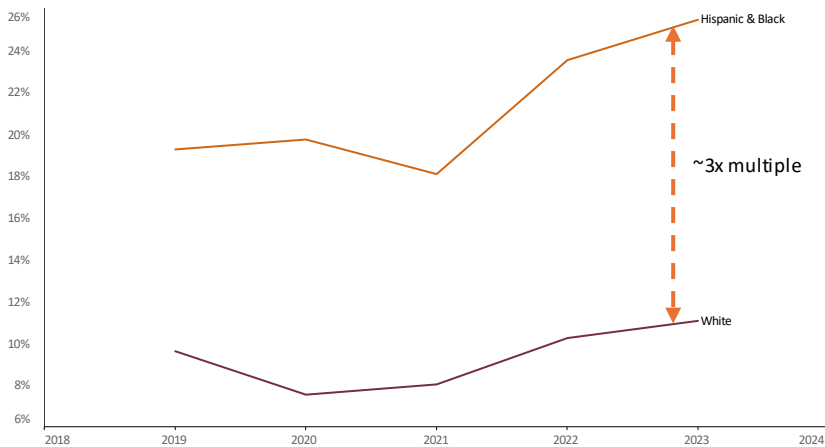


| Money is the Real Problem

Hispanic and Black households experience food insecurity at 2x the rate of White households. But when normalized for income, the gap narrows sharply, revealing that the racial food divide is largely a reflection of an income divide that has barely budged in over two decades.

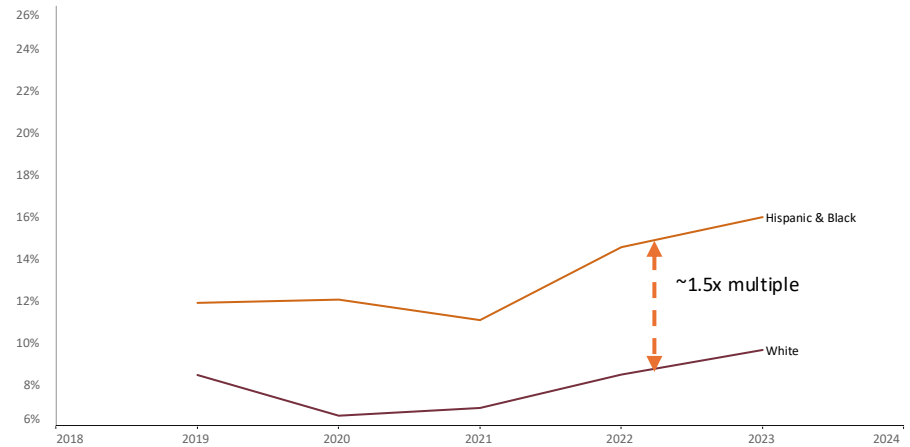
Food scarcity is ~3x worse for Hispanic and Black populations...

Rate of Food Insecurity



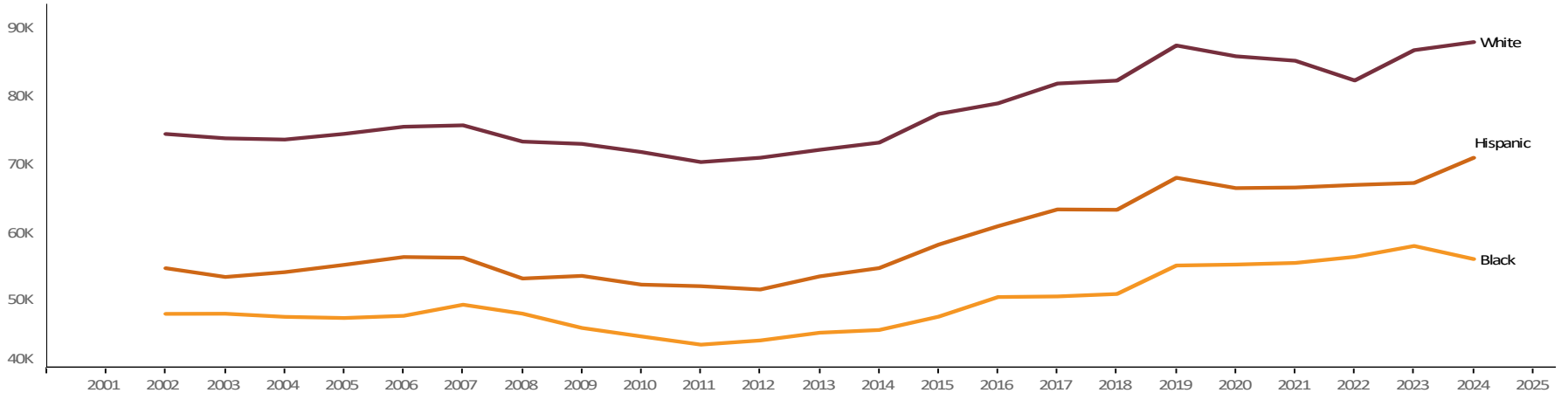
...but this gap reduces when we normalize based on median income

Rate of Food Insecurity Normalized by Median Income



Income disparity amongst racial groups has remained extremely stubborn for the past 20 years

Median Income (\$)



| Poor Communities Struggle to Buy Healthy Food

Food insecurity is not randomly distributed – it tracks poverty closely. Every increasing in a county's poverty rate corresponds to a measurable rise in food insecurity, making them two faces of the same underlying problem

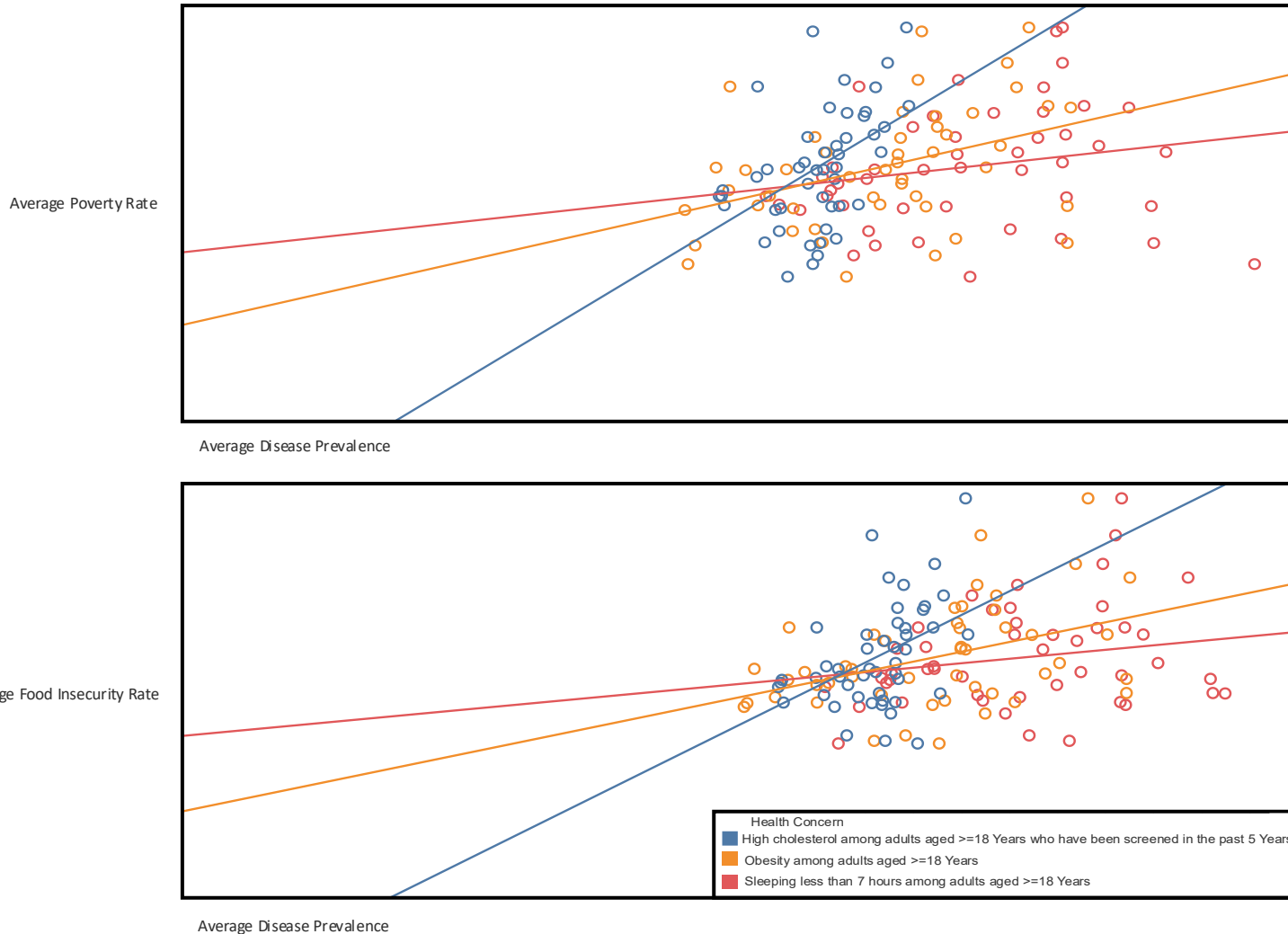
Food insecurity increases with poverty



| When You Can't Eat Well, You Get Sicker

Each health concern, obesity, high cholesterol, and less sleep all follow the same pattern: counties with the highest poverty and food insecurity rates show consistently worse outcomes, suggesting these conditions do not just coexist with poor health – they shape it.

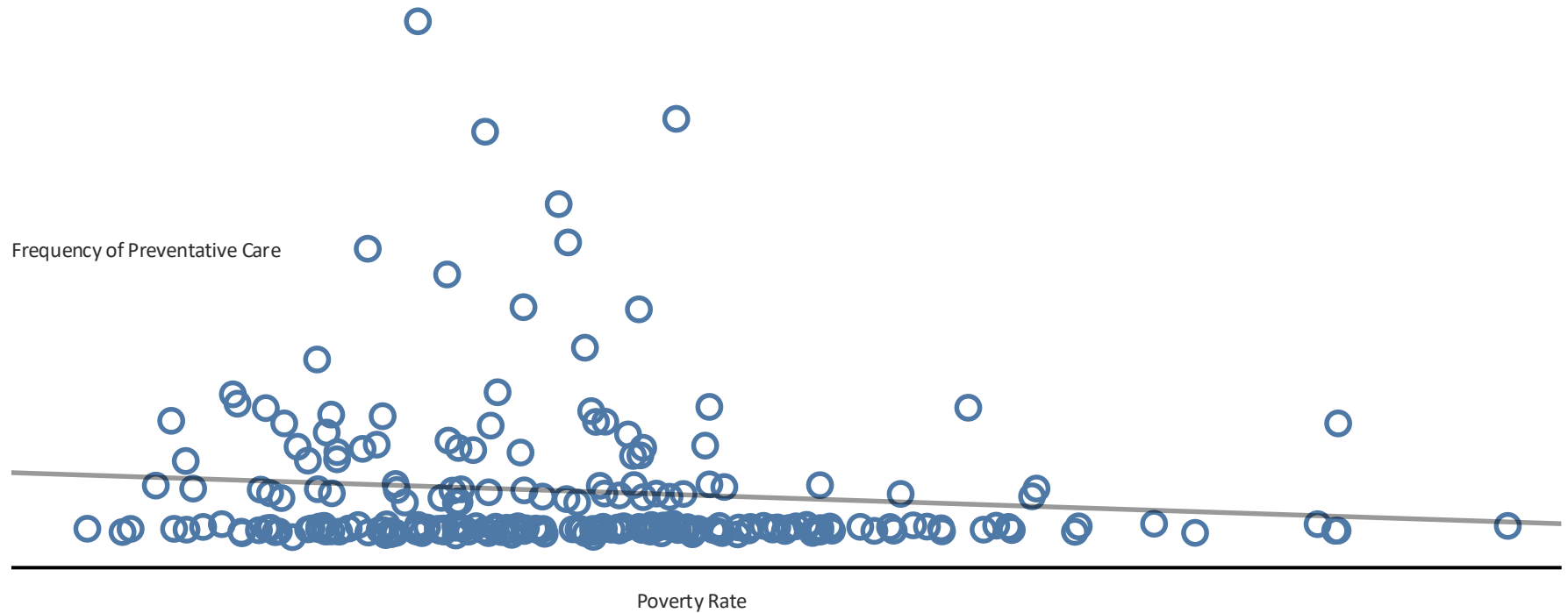
Prevalence of disease increases with higher poverty and food insecurity rates



| Access to Preventive Care Works in the Opposite Direction

Preventive care follows the opposite trajectory of poverty. Where economic hardship is highest, regular doctor visits drop off, leaving treatable conditions to progress unmonitored and uncared.

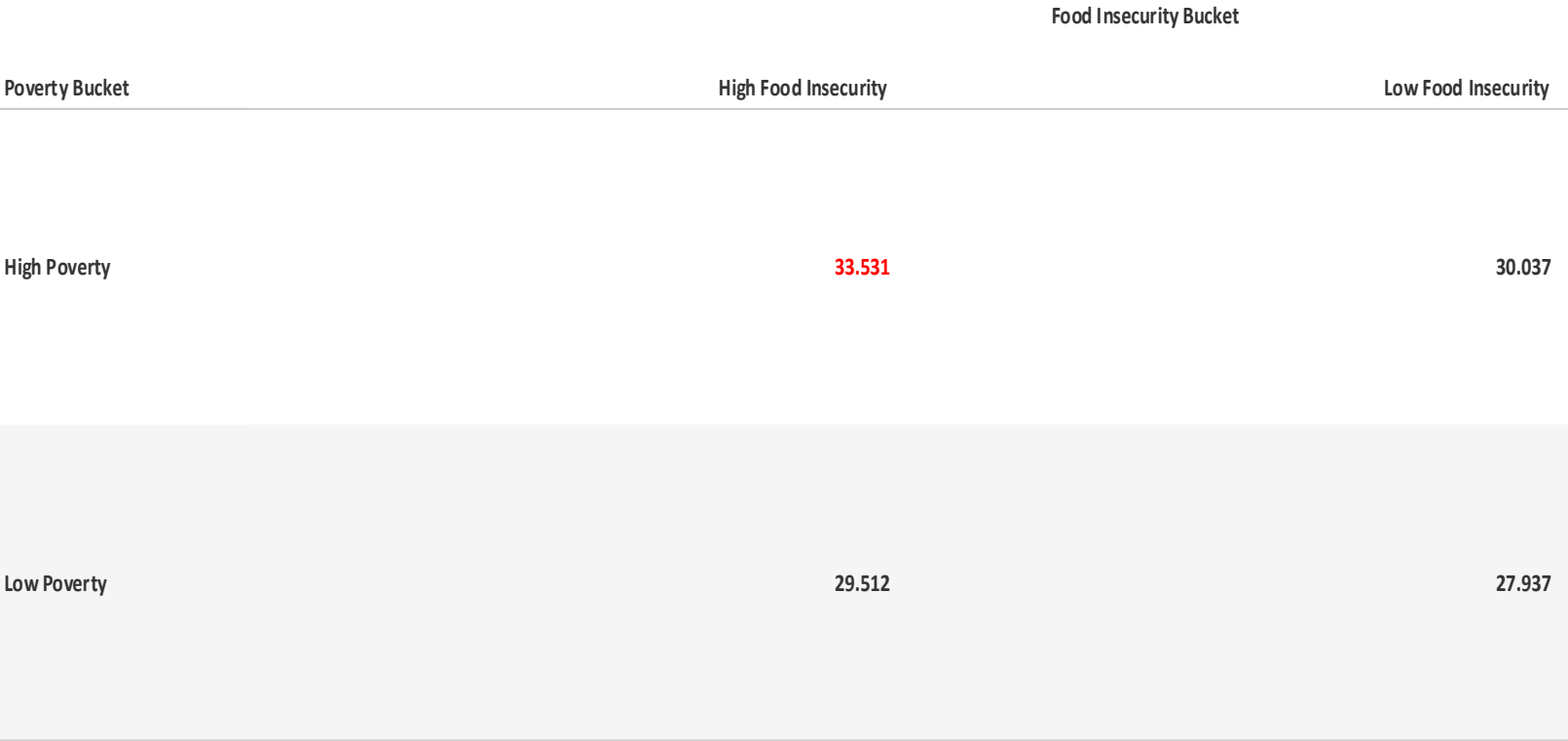
Quality of preventative care declines as poverty rate rises



Low Income and Food Access Have a Compounding Effect

Splitting counties by national medians on poverty and food insecurity reveals a clear pattern: the 'high-both' quadrant consistently shows the highest proportion of adverse health outcomes, and the gap widens beyond what either factor predicts independently.

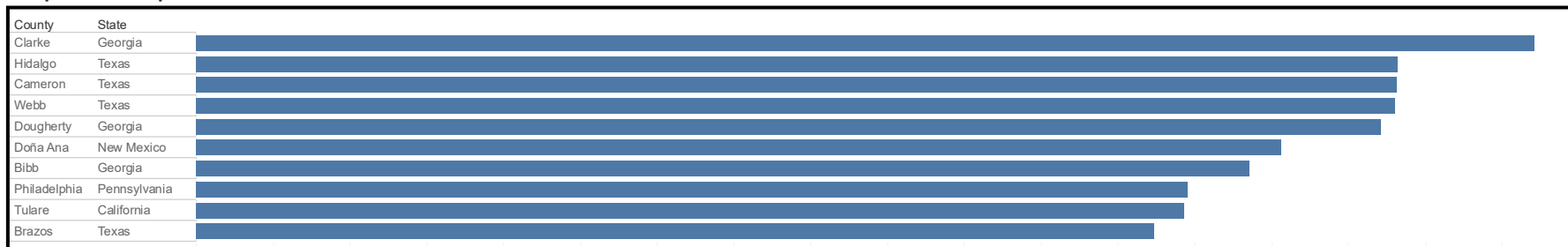
Counties Facing Both Crises See the Worst Health Outcomes



The Worst Counties Repeatedly Appear Across Multiple Metrics

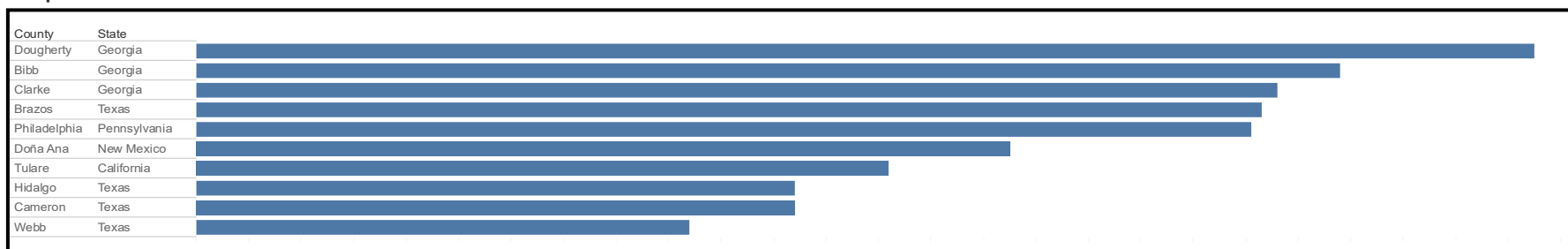
The same counties appear again and again: highest poverty, highest food insecurity and worst health outcomes. These are not separate crises happening in separate places and the overlap isn't coincidence – it's the pattern this entire analysis has been describing.

Top 10 most impoverished counties



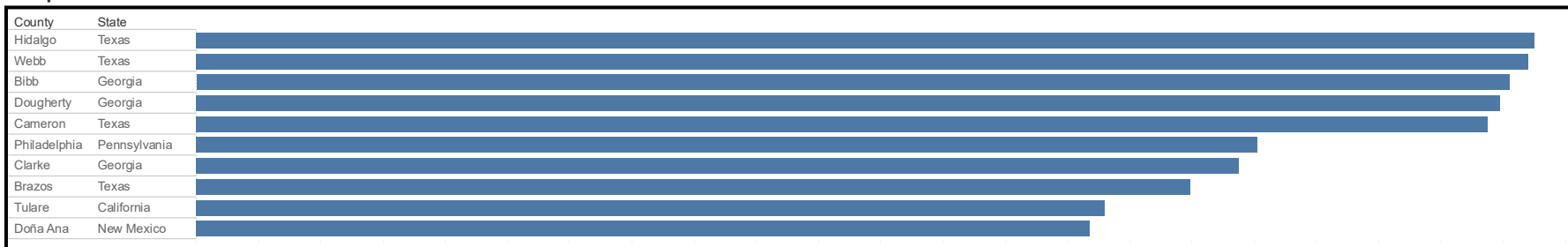
Average Poverty Rate

Top 10 most food insecure counties



Average Food Insecurity Rate

Top 10 most health concerned counties



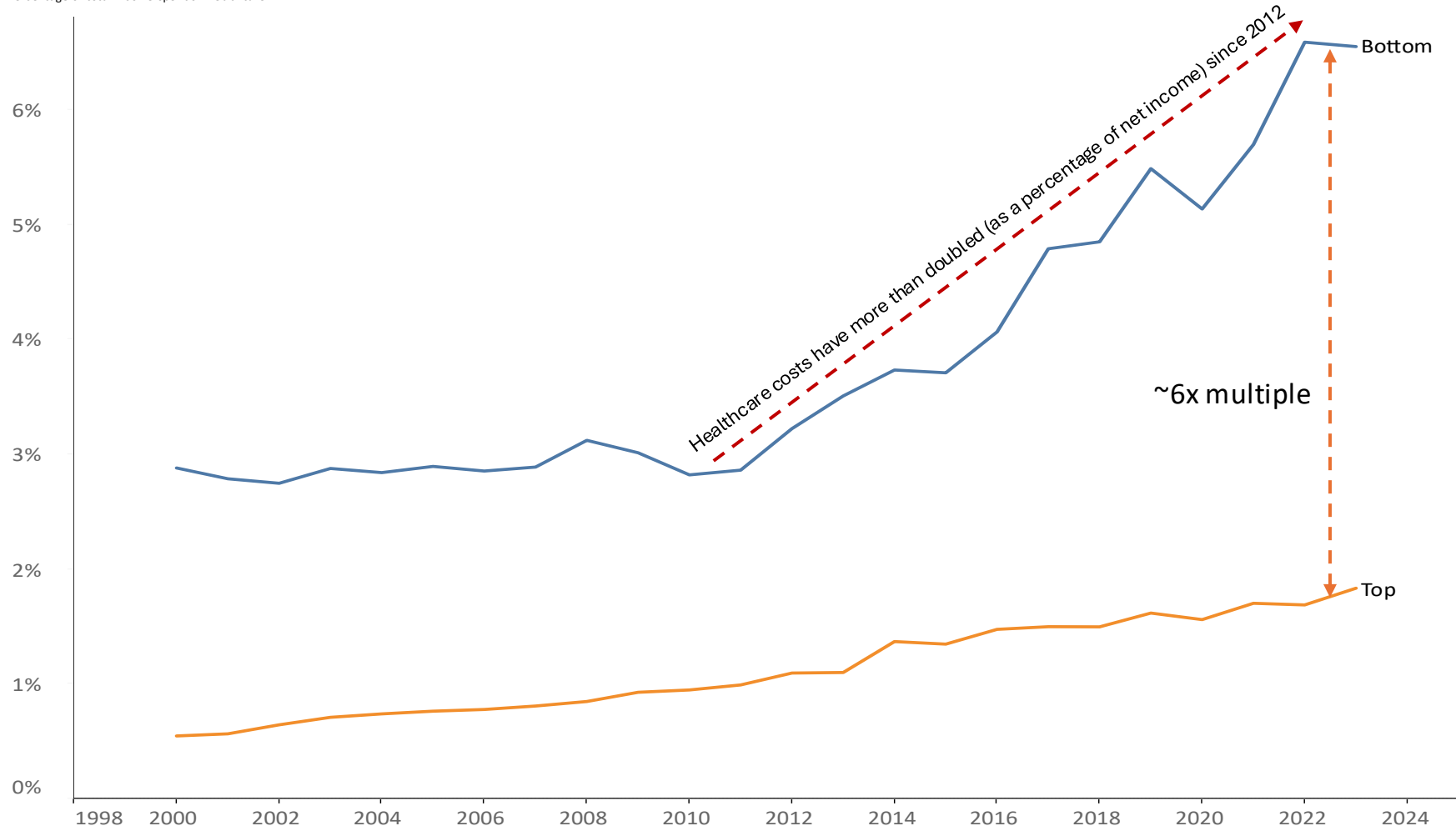
Average Disease Prevalence

| Expensive Healthcare Pushes Families Deeper into Poverty

As individuals in the lowest income brackets lose access to food and preventative care, they develop costly health crises that consume an ever-larger share of their meager income. This forces deeper cuts to nutrition and healthcare which worsens health and triggers further healthcare expenses and the vicious cycle repeats.

As a percentage of net income, healthcare costs are over 6x more for households in the bottom quintile of incomes

Percentage of total income spent on healthcare



Key Takeaways & Conclusions

- 1 Food insecurity follows poverty and race. It is not random.
- 2 The crisis is concentrated in specific regions, especially the South
- 3 Poverty and hunger creates a trap: bad nutrition leads to worse health
- 4 Medical bills hit poor families the hardest and make things worse
- 5 Focused support in the hardest-hit counties could help break this cycle

☐ **Bottom line: poor health outcomes amongst low-income households could explain why people fail to free themselves from poverty and food insecurity**